# SOLID MECHANICS FOR IBFE

Kinematics, strain, stress, and hyperelasticity

$$\chi \longrightarrow \boldsymbol{F}_\chi \longrightarrow W \longrightarrow \boldsymbol{P}$$

motion, deformation, energy, stress

Laura A. Miller

Department of Mathematics, University of Arizona

IBFE Tutorials • Module 4

AI tools were used to assist with formatting, organization, and clarification of the instructor's original notes.

# ROADMAP

1. Notation and tensor bookkeeping
2. Configurations and the deformation map
3. Finite strain measures
4. Stress measures and transformations
5. Hyperelasticity and the IBFE force pipeline
6. Worked axial-deformation example
7. Neo-Hookean stress derivation and checks

$$\chi \longrightarrow \boldsymbol{F}_{\chi}, J \longrightarrow \boldsymbol{C}, \boldsymbol{E} \longrightarrow W \longrightarrow \boldsymbol{P} \longrightarrow \boldsymbol{F} \longrightarrow \boldsymbol{f}.$$

The continuum-mechanics material in this deck is standard and is adapted from Holzapfel (2000) and Bonet & Wood (2008); the IBFE force construction follows Griffith & Luo (2017) and Griffith & Patankar (2020). Traction figures are from P. A. Kelly's Auckland lecture notes (credited on each slide). Full references are on the final slide.

# MODULE 0: PURPOSE AND SCOPE

**Goal**
Build the minimum solid-mechanics toolkit needed to follow IBFE derivations.

**Audience**
Advanced undergraduates and graduate students who are new to continuum mechanics.

This module establishes:

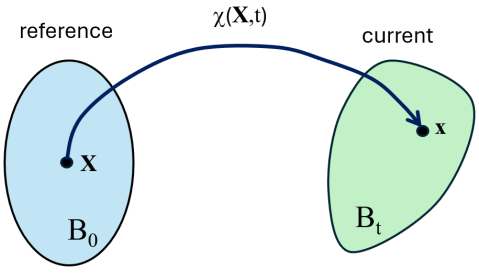
- reference and current coordinates,
- tensor types and indices,
- material and spatial derivatives,
- distinct symbols for deformation and force.

# CONFIGURATIONS, COORDINATES, AND FIELDS

- Reference configuration: $B_0 \subset \mathbb{R}^d$, with material coordinate $\boldsymbol{X}$.
- Current configuration: $B_t \subset \mathbb{R}^d$, with spatial coordinate $\boldsymbol{x}$.
- Motion: $\chi(\cdot, t) : B_0 \to B_t$, with

$$\boldsymbol{x} = \chi(\boldsymbol{X}, t).$$

Material fields use $\boldsymbol{X}$ as their independent coordinate. Spatial fields use $\boldsymbol{x}$.




A material point keeps its label $\boldsymbol{X}$ while its current position changes.

Schematic: instructor's course notes.

# SCALARS, VECTORS, AND SECOND-ORDER TENSORS

| Object | Components | Example |
|---|---|---|
| Scalar | $a$ | energy density $W$ |
| Vector | $v_i$ or $v_I$ | velocity, normal, force |
| Spatial tensor | $A_{ij}$ | Cauchy stress $\sigma_{ij}$ |
| Material tensor | $A_{IJ}$ | Green strain $E_{IJ}$ |
| Two-point tensor | $A_{iI}$ | $(F_\chi)_{iI}$, $P_{iI}$ |

A two-point tensor maps a material direction into a spatial direction. Both the deformation gradient and first Piola stress have one spatial and one material index.

$$i, j, k: \text{ spatial}, \qquad I, J, K: \text{ material}.$$

Repeated indices are summed unless stated otherwise.

# MATERIAL AND SPATIAL FIELDS

| Reference domain $B_0$ | Current domain $B_t$ or fluid domain $\Omega$ |
|---|---|
| $\boldsymbol{X}$: material coordinate | $\boldsymbol{x}$: spatial coordinate |
| $\chi(\boldsymbol{X}, t)$: motion | $\boldsymbol{u}(\boldsymbol{x}, t)$: velocity |
| $\boldsymbol{F}_\chi = \nabla_X \chi$: deformation gradient | $\boldsymbol{\sigma}(\boldsymbol{x}, t)$: Cauchy stress |
| $J = \det \boldsymbol{F}_\chi$: volume ratio | $p(\boldsymbol{x}, t)$: pressure |
| $\boldsymbol{P}(\boldsymbol{X}, t)$: first Piola stress | $\boldsymbol{f}(\boldsymbol{x}, t)$: Eulerian force density |
| $\boldsymbol{F}(\boldsymbol{X}, t)$: Lagrangian force density | |

> Always check the independent coordinate before differentiating or integrating a field.

# MATERIAL AND SPATIAL DERIVATIVES

**Material gradient**

$$\nabla_X(\cdot)$$

differentiates with respect to $\boldsymbol{X}$. For the motion,

$$(\nabla_X \chi)_{iI} = \frac{\partial \chi_i}{\partial X_I}.$$

**Spatial gradient**

$$\nabla_x(\cdot)$$

differentiates with respect to $\boldsymbol{x}$. For velocity,

$$(\nabla_x \boldsymbol{u})_{ij} = \frac{\partial u_i}{\partial x_j}.$$

IBFE structural kinematics and constitutive stress are normally evaluated on $B_0$, so $\nabla_X$ appears throughout the finite-element formulation.

# DEFORMATION GRADIENT AND LAGRANGIAN FORCE

IB and IBFE papers often use bold $F$ for a Lagrangian force density, while solid-mechanics texts use bold $F$ for the deformation gradient.

| **Kinematics** | **Forcing** |
| --- | --- |
| $\boldsymbol{F}_\chi(\boldsymbol{X},t) := \nabla_X \chi(\boldsymbol{X},t)$ | $\boldsymbol{F}(\boldsymbol{X},t)$ |
| is a two-point tensor. | is a Lagrangian vector force density. |

$$\boldsymbol{F}_\chi \neq \boldsymbol{F}.$$

The subscript on $\boldsymbol{F}_\chi$ keeps kinematics distinct from forcing.

# MODULE 1: CONFIGURATIONS AND MOTION

$$\chi(\cdot, t) : B_0 \to B_t, \qquad \boldsymbol{x} = \chi(\boldsymbol{X}, t).$$

The displacement is

$$\boldsymbol{d}(\boldsymbol{X}, t) = \chi(\boldsymbol{X}, t) - \boldsymbol{X}.$$

This module develops two local measures from the motion:

$$\boldsymbol{F}_\chi = \nabla_X \chi, \qquad J = \det \boldsymbol{F}_\chi.$$

$\boldsymbol{F}_\chi$ maps line elements and $J$ maps volume elements.

# NEARBY MATERIAL POINTS UNDER THE MOTION

Choose neighboring material points

$$\boldsymbol{X}, \qquad \boldsymbol{X} + d\boldsymbol{X}.$$

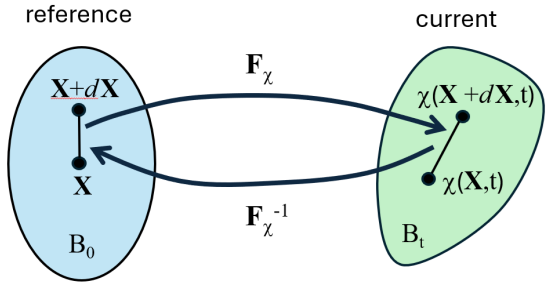
Their current positions are

$$\boldsymbol{x} = \chi(\boldsymbol{X}, t), \qquad \boldsymbol{x} + d\boldsymbol{x} = \chi(\boldsymbol{X} + d\boldsymbol{X}, t).$$

Thus

$$d\boldsymbol{x} = \chi(\boldsymbol{X} + d\boldsymbol{X}, t) - \chi(\boldsymbol{X}, t).$$

> The deformation gradient is the linear map that approximates this change for small $d\boldsymbol{X}$.




The arrows labelled $\boldsymbol{F}_\chi$ and $\boldsymbol{F}_\chi^{-1}$ act on the short line element joining the two points, not on the points themselves.

Schematic: instructor's course notes.

# DEFORMATION-GRADIENT DEFINITION

Linearizing the motion about $\boldsymbol{X}$ gives

$$d\boldsymbol{x} = \boldsymbol{F}_\chi(\boldsymbol{X}, t)\, d\boldsymbol{X} + o(\|d\boldsymbol{X}\|),$$

with

$$\boxed{\boldsymbol{F}_\chi(\boldsymbol{X}, t) = \nabla_X \chi(\boldsymbol{X}, t)}.$$

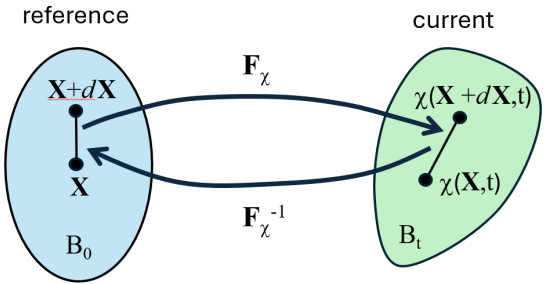
In components,

$$(F_\chi)_{iI} = \frac{\partial \chi_i}{\partial X_I}.$$

**Columns of $\boldsymbol{F}_\chi$**
Images of the reference basis vectors.

**Information in $\boldsymbol{F}_\chi$**
Stretch, shear, and rotation near one material point.

# INVERSE DIFFERENTIAL MAP AT MATCHING POINTS




Schematic: instructor's course notes.

If $\chi(\cdot, t)$ is locally invertible and $\boldsymbol{x} = \chi(\boldsymbol{X}, t)$, then

$$d\boldsymbol{X} = \boldsymbol{F}_\chi(\boldsymbol{X}, t)^{-1} d\boldsymbol{x} + o(\|d\boldsymbol{x}\|).$$

The spatial gradient of the inverse motion is

$$\boxed{\nabla_x \chi^{-1}(\boldsymbol{x}, t) = \boldsymbol{F}_\chi(\chi^{-1}(\boldsymbol{x}, t), t)^{-1}}.$$

> The composition with $\chi^{-1}$ ensures that both tensors refer to the same material point.

# JACOBIAN AND LOCAL VOLUME CHANGE

$$\boxed{J(\boldsymbol{X},t) = \det \boldsymbol{F}_\chi(\boldsymbol{X},t)}, \qquad dv = J\, dV.$$

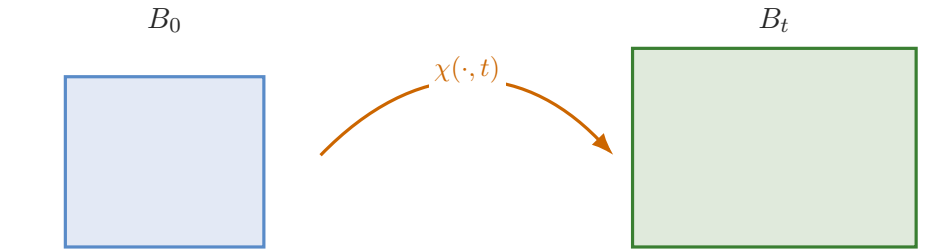
- $J > 1$: local expansion
- $0 < J < 1$: local compression
- $J = 1$: local volume preservation

An admissible deformation normally requires

$$J > 0.$$

$J = 0$ collapses material volume. $J < 0$ reverses orientation and indicates element inversion.

# VISUALIZING THE JACOBIAN AS A LOCAL MEASURE RATIO




$$dv = J\,dV, \qquad J = \det \boldsymbol{F}_\chi$$

In two dimensions the corresponding area relation is $da = J\,dA$.

# JACOBIAN AS THE PRODUCT OF PRINCIPAL STRETCHES

For a diagonal deformation in $d$ dimensions,

$$\boldsymbol{F}_\chi = \operatorname{diag}(\lambda_1, \ldots, \lambda_d), \qquad J = \prod_{k=1}^{d} \lambda_k.$$

**Two dimensions**

$$da = J\, dA = \lambda_1 \lambda_2\, dA.$$

**Three dimensions**

$$dv = J\, dV = \lambda_1 \lambda_2 \lambda_3\, dV.$$

> Volume preservation does not imply zero strain. A simple shear has $J = 1$ and still changes angles and lengths.

# MICRO-CHECK: DEFORMATION GRADIENTS AND JACOBIANS

## A. Uniform stretch

$$\chi(\boldsymbol{X}) = (\lambda X_1, \lambda X_2, \lambda X_3).$$

1. Compute $\boldsymbol{F}_\chi = \nabla_X \chi$.
2. Compute $J = \det \boldsymbol{F}_\chi$.
3. Interpret the volume change.

## B. Simple shear

$$\chi(\boldsymbol{X}) = (X_1 + \gamma X_2, X_2, X_3).$$

1. Compute each $(F_\chi)_{iI}$.
2. Compute $J$.
3. Decide whether the map is strain free.

# MICRO-CHECK SOLUTION: UNIFORM STRETCH

From

$$\chi_1 = \lambda X_1, \qquad \chi_2 = \lambda X_2, \qquad \chi_3 = \lambda X_3,$$

we obtain

$$\boldsymbol{F}_\chi = \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} = \lambda \boldsymbol{I}.$$

Therefore

$$\boxed{J = \det \boldsymbol{F}_\chi = \lambda^3}.$$

> Every length scales by $\lambda$, every area by $\lambda^2$, and every volume by $\lambda^3$.

# MICRO-CHECK SOLUTION: SIMPLE SHEAR

For

$$\chi(\boldsymbol{X}) = (X_1 + \gamma X_2, X_2, X_3),$$

$$\boldsymbol{F}_\chi = \begin{bmatrix} 1 & \gamma & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Because this matrix is upper triangular with unit diagonal entries, its determinant is the product of those entries:

$$\boxed{J = 1}.$$

> Simple shear preserves volume but changes line lengths and angles. The strain measures in Module 2 detect those changes.

# THE DISPLACEMENT GRADIENT IS NOT A FINITE STRAIN

Since

$$\chi(\boldsymbol{X}, t) = \boldsymbol{X} + \boldsymbol{d}(\boldsymbol{X}, t),$$

we have

$$\nabla_X \boldsymbol{d} = \boldsymbol{F}_\chi - \boldsymbol{I}.$$

**What it is**

A first derivative of displacement that contains stretch, shear, and rotation.

**Small-strain approximation**

$$\boldsymbol{\varepsilon} = \frac{1}{2}\left(\nabla_X \boldsymbol{d} + (\nabla_X \boldsymbol{d})^T\right).$$

> Finite deformation requires nonlinear strain measures that remain unchanged under rigid-body motion.

# MODULE 1 WRAP-UP

- The motion maps material points: $\boldsymbol{x} = \chi(\boldsymbol{X}, t)$.
- The deformation gradient maps line elements:

$$\boldsymbol{F}_\chi = \nabla_X \chi, \qquad d\boldsymbol{x} \approx \boldsymbol{F}_\chi \, d\boldsymbol{X}.$$

- The Jacobian maps volume elements:

$$J = \det \boldsymbol{F}_\chi, \qquad dv = J \, dV.$$

- The material and spatial gradients act on different independent coordinates.

> Quick check: if $\chi(\boldsymbol{X}) = \lambda \boldsymbol{X}$ in three dimensions, then $\boldsymbol{F}_\chi = \lambda \boldsymbol{I}$ and $J = \lambda^3$.

# MODULE 2: REQUIREMENTS FOR A FINITE STRAIN

A useful finite-strain measure must:

- detect stretch and shear,
- ignore rigid translation and rigid rotation,
- remain meaningful for large deformation,
- reduce to linearized strain when $\|\nabla_X \boldsymbol{d}\| \ll 1$.

> The deformation gradient contains both deformation and rotation. Strain tensors built from $\boldsymbol{F}_\chi^T \boldsymbol{F}_\chi$ or $\boldsymbol{F}_\chi \boldsymbol{F}_\chi^T$ separate those effects.

# ROTATION AND STRETCH SHOULD FEEL DIFFERENT

**Pure rotation**

$$\boldsymbol{F}_\chi = \boldsymbol{Q}, \qquad \boldsymbol{Q}^T\boldsymbol{Q} = \boldsymbol{I}.$$

Lengths and angles remain unchanged.

desired strain $= \mathbf{0}$.

**Pure stretch**

$$\boldsymbol{F}_\chi = \mathrm{diag}(\lambda_1, \lambda_2, \lambda_3).$$

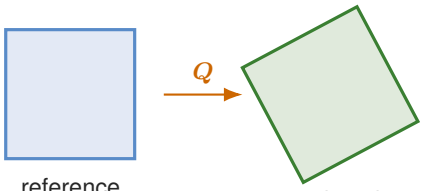
At least one length changes when $\lambda_i \neq 1$.

desired strain $\neq \mathbf{0}$.

> This test immediately rules out $\boldsymbol{F}_\chi - \boldsymbol{I}$: a finite rotation gives $\boldsymbol{Q} - \boldsymbol{I} \neq \mathbf{0}$.

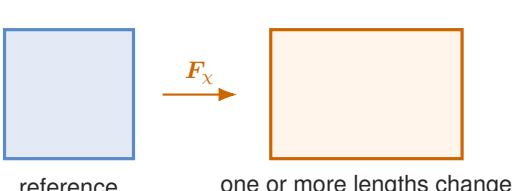
# VISUAL CHECK: RIGID ROTATION VERSUS GENUINE STRETCH

## Pure rotation




$\boldsymbol{Q}^T\boldsymbol{Q} = \boldsymbol{I} \quad \Rightarrow \quad \text{strain} = \mathbf{0}$

## Pure stretch




$\lambda_i \neq 1 \quad \Rightarrow \quad \text{strain} \neq \mathbf{0}$

# DISPLACEMENT GRADIENT AND LINEARIZED STRAIN

With

$$\chi = \boldsymbol{X} + \boldsymbol{d}, \qquad \boldsymbol{F}_\chi = \boldsymbol{I} + \nabla_X \boldsymbol{d},$$

the linearized strain is

$$\boxed{\boldsymbol{\varepsilon} = \mathrm{sym}(\nabla_X \boldsymbol{d}) = \frac{1}{2}\left(\nabla_X \boldsymbol{d} + (\nabla_X \boldsymbol{d})^T\right).}$$

> This approximation assumes small displacement gradients, including small rotations. Finite rotations require a nonlinear strain measure.

# RIGHT AND LEFT CAUCHY–GREEN TENSORS

$$\boldsymbol{C} = \boldsymbol{F}_\chi^T \boldsymbol{F}_\chi \qquad \text{(material)}$$

$$\boldsymbol{B} = \boldsymbol{F}_\chi \boldsymbol{F}_\chi^T \qquad \text{(spatial)}$$

When $J > 0$, both are symmetric positive definite and share the eigenvalues $\lambda_i^2$.

Under a superposed spatial rotation $\boldsymbol{F}_\chi^\star = \boldsymbol{Q}\boldsymbol{F}_\chi$,

$$\boldsymbol{C}^\star = \boldsymbol{C}, \qquad \boldsymbol{B}^\star = \boldsymbol{Q}\boldsymbol{B}\boldsymbol{Q}^T.$$

$\boldsymbol{C}$ remains unchanged. $\boldsymbol{B}$ rotates with the spatial frame, while its scalar invariants remain unchanged.

# GREEN–LAGRANGE STRAIN

$$\boxed{\boldsymbol{E} = \frac{1}{2}(\boldsymbol{C} - \boldsymbol{I}) = \frac{1}{2}(\boldsymbol{F}_\chi^T \boldsymbol{F}_\chi - \boldsymbol{I})}.$$

In components,

$$E_{IJ} = \frac{1}{2}(C_{IJ} - \delta_{IJ}), \qquad C_{IJ} = \sum_{i=1}^{d} (F_\chi)_{iI}(F_\chi)_{iJ}.$$

**Configuration**
$\boldsymbol{E}$ acts on material line elements.

**Small-deformation limit**
$\boldsymbol{E} = \boldsymbol{\varepsilon} + O(\|\nabla_X \boldsymbol{d}\|^2)$.

# GREEN–LAGRANGE STRAIN AND LENGTH CHANGE

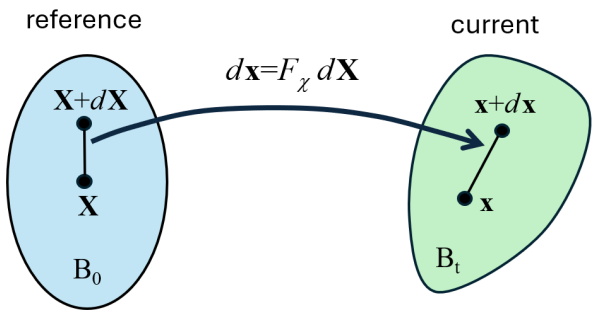
For a material line element,

$$d\boldsymbol{x} = \boldsymbol{F}_\chi \, d\boldsymbol{X}.$$

Then

$$\|d\boldsymbol{x}\|^2 - \|d\boldsymbol{X}\|^2 = 2\, d\boldsymbol{X} \cdot \boldsymbol{E}\, d\boldsymbol{X}.$$

> $\boldsymbol{E} = \boldsymbol{0}$ means that local lengths are preserved. For a compatible deformation of a connected body, this corresponds to rigid motion.




$$|d\boldsymbol{x}|^2 - |d\boldsymbol{X}|^2 \;=\; 2\, d\boldsymbol{X} \cdot \boldsymbol{E}\, d\boldsymbol{X}$$

The strain is expressed using the original material line element.

Schematic: instructor's course notes.

# EULER–ALMANSI STRAIN

The spatial finite-strain measure is

$$\boxed{\boldsymbol{e} = \frac{1}{2}(\boldsymbol{I} - \boldsymbol{B}^{-1}) = \frac{1}{2}\left[\boldsymbol{I} - (\boldsymbol{F}_\chi \boldsymbol{F}_\chi^T)^{-1}\right]}.$$

**Green–Lagrange**
$\boldsymbol{E}$ acts on vectors in the reference configuration.

**Euler–Almansi**
$\boldsymbol{e}$ acts on vectors in the current configuration.

> IBFE derivations usually favor $\boldsymbol{F}_\chi$, $\boldsymbol{C}$, and $\boldsymbol{E}$ because finite-element integration occurs on $B_0$.

# EULER–ALMANSI STRAIN AND CURRENT LINE ELEMENTS

Take a current line element $d\boldsymbol{x}$. Its material counterpart is

$$d\boldsymbol{X} = \boldsymbol{F}_\chi^{-1} d\boldsymbol{x}.$$

Therefore

$$\|d\boldsymbol{x}\|^2 - \|d\boldsymbol{X}\|^2 = d\boldsymbol{x} \cdot (\boldsymbol{I} - \boldsymbol{B}^{-1}) d\boldsymbol{x} = 2\, d\boldsymbol{x} \cdot \boldsymbol{e}\, d\boldsymbol{x}.$$

The two finite-strain measures are related by

$$\boxed{\boldsymbol{e} = \boldsymbol{F}_\chi^{-T} \boldsymbol{E} \boldsymbol{F}_\chi^{-1}}.$$

> The formulas describe the same deformation using different coordinates.

# MICRO-CHECK: PURE ROTATION

Let

$$\chi(\boldsymbol{X}) = \boldsymbol{Q}\boldsymbol{X} + \boldsymbol{c}, \qquad \boldsymbol{Q}^T\boldsymbol{Q} = \boldsymbol{I}, \qquad \det \boldsymbol{Q} = 1.$$

Then

$$\boldsymbol{F}_\chi = \boldsymbol{Q}, \qquad \boldsymbol{C} = \boldsymbol{Q}^T\boldsymbol{Q} = \boldsymbol{I}, \qquad \boldsymbol{E} = \boldsymbol{0}.$$

Also,

$$\boldsymbol{B} = \boldsymbol{Q}\boldsymbol{Q}^T = \boldsymbol{I}, \qquad \boldsymbol{e} = \boldsymbol{0}.$$

> Both finite-strain measures pass the rigid-rotation test.

# MICRO-CHECK: SIMPLE SHEAR

For

$$\boldsymbol{F}_\chi = \begin{bmatrix} 1 & \gamma & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

compute

$$\boldsymbol{C} = \begin{bmatrix} 1 & \gamma & 0 \\ \gamma & 1+\gamma^2 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad \boldsymbol{E} = \begin{bmatrix} 0 & \gamma/2 & 0 \\ \gamma/2 & \gamma^2/2 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

> Linearized strain retains the terms proportional to $\gamma$. Finite strain also contains the normal-strain term $\gamma^2/2$.

# POLAR DECOMPOSITION: $\boldsymbol{F}_\chi = \boldsymbol{R}\boldsymbol{U} = \boldsymbol{V}\boldsymbol{R}$

Every deformation gradient with $\det \boldsymbol{F}_\chi > 0$ factors uniquely as

$$\boxed{\boldsymbol{F}_\chi = \boldsymbol{R}\boldsymbol{U} = \boldsymbol{V}\boldsymbol{R}}, \qquad \boldsymbol{R}^T\boldsymbol{R} = \boldsymbol{I}, \qquad \det \boldsymbol{R} = 1,$$

$$\boldsymbol{U} = \sqrt{\boldsymbol{F}_\chi^T \boldsymbol{F}_\chi} = \sqrt{\boldsymbol{C}}, \qquad \boldsymbol{V} = \sqrt{\boldsymbol{F}_\chi \boldsymbol{F}_\chi^T} = \sqrt{\boldsymbol{B}}, \qquad \boldsymbol{V} = \boldsymbol{R}\boldsymbol{U}\boldsymbol{R}^T.$$

The right stretch $\boldsymbol{U}$ and the left stretch $\boldsymbol{V}$ are symmetric positive definite; $\boldsymbol{R}$ is proper orthogonal.

**Reading $\boldsymbol{R}\boldsymbol{U}$**
Stretch first along the eigenvectors of $\boldsymbol{U}$ by the principal stretches $\lambda_i$, then rotate by $\boldsymbol{R}$.

**Reading $\boldsymbol{V}\boldsymbol{R}$**
Rotate first by the same $\boldsymbol{R}$, then stretch by the same $\lambda_i$ in the rotated frame.

The eigenvalues of $\boldsymbol{U}$ are the principal stretches $\lambda_i$; the eigenvalues of $\boldsymbol{C} = \boldsymbol{U}^2$ are $\lambda_i^2$.

> Consequence. Objectivity demands $W(\boldsymbol{Q}\boldsymbol{F}_\chi) = W(\boldsymbol{F}_\chi)$ for every rotation $\boldsymbol{Q}$. Take $\boldsymbol{Q} = \boldsymbol{R}^T$: $W(\boldsymbol{F}_\chi) = W(\boldsymbol{R}^T\boldsymbol{R}\boldsymbol{U}) = W(\boldsymbol{U}) =: \widehat{W}(\boldsymbol{U}) = \widetilde{W}(\boldsymbol{C})$. A strain energy can see $\boldsymbol{F}_\chi$ only through $\boldsymbol{U}$, equivalently through $\boldsymbol{C} = \boldsymbol{U}^2$.

# POLAR DECOMPOSITION: WORKED CHECK

Take the simple shear of the previous slide in the $X_1X_2$ plane with $\gamma = 1$:

$$\boldsymbol{F}_\chi = \begin{bmatrix} 1 & \gamma \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \qquad \boldsymbol{C} = \boldsymbol{F}_\chi^T \boldsymbol{F}_\chi = \begin{bmatrix} 1 & \gamma \\ \gamma & 1+\gamma^2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}.$$

**Principal stretches.** $\lambda_{1,2} = \frac{1}{2}\left(\sqrt{4+\gamma^2} \pm \gamma\right)$, so $\lambda_1 = \frac{\sqrt{5}+1}{2} \approx 1.618$, $\lambda_2 = \frac{\sqrt{5}-1}{2} \approx 0.618$, and $\lambda_1\lambda_2 = 1 = \det \boldsymbol{F}_\chi$. The eigenvalues of $\boldsymbol{C}$ are $\lambda_1^2 \approx 2.618$ and $\lambda_2^2 \approx 0.382$.

$$\boldsymbol{U} = \sqrt{\boldsymbol{C}} = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \approx \begin{bmatrix} 0.894 & 0.447 \\ 0.447 & 1.342 \end{bmatrix}, \qquad \boldsymbol{R} = \boldsymbol{F}_\chi \boldsymbol{U}^{-1} = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & 1 \\ -1 & 2 \end{bmatrix} \approx \begin{bmatrix} 0.894 & 0.447 \\ -0.447 & 0.894 \end{bmatrix}.$$

Checks: $\boldsymbol{U}^2 = \boldsymbol{C}$, $\boldsymbol{R}^T\boldsymbol{R} = \boldsymbol{I}$, $\det \boldsymbol{R} = 1$, and $\boldsymbol{R}$ is a clockwise rotation through $\theta$ with $\tan\theta = \gamma/2 = \frac{1}{2}$, that is $\theta \approx 26.6^\circ$.

> Simple shear is a stretch of $\lambda_1$ along one principal direction, a compression $\lambda_2 = 1/\lambda_1$ along the other, plus a rigid rotation by $\theta$; that rotation is exactly what $\boldsymbol{C}$ and $\boldsymbol{E}$ discard.

# POLAR DECOMPOSITION: LEFT STRETCH, ROTATION SENSE

For the same $\gamma = 1$ shear, the left stretch follows from $\boldsymbol{V} = \boldsymbol{F}_\chi \boldsymbol{R}^T$:

$$\boldsymbol{V} = \frac{1}{\sqrt{5}} \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix} \approx \begin{bmatrix} 1.342 & 0.447 \\ 0.447 & 0.894 \end{bmatrix}, \qquad \boldsymbol{V}^2 = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} = \boldsymbol{F}_\chi \boldsymbol{F}_\chi^T = \boldsymbol{B}.$$

**Principal directions**

Eigenvectors of $\boldsymbol{U}$ (reference): $\boldsymbol{N}_1 \propto (1, \lambda_1)$, about $58.3°$ from $X_1$; $\boldsymbol{N}_2 \propto (-\lambda_1, 1)$.
Eigenvectors of $\boldsymbol{V}$ (current): $\boldsymbol{n}_i = \boldsymbol{R}\boldsymbol{N}_i$, so $\boldsymbol{n}_1$ lies about $58.3° - 26.6° = 31.7°$ from $x_1$.

**Why "clockwise"**

Write $\boldsymbol{R} = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$ with $\cos\theta = 2/\sqrt{5}$, $\sin\theta = 1/\sqrt{5}$. Then $\boldsymbol{R}\boldsymbol{e}_1 = (2, -1)/\sqrt{5}$ lies *below* the $x_1$ axis: a rotation through $-\theta$, i.e. clockwise for $x_1$ to the right and $x_2$ up. For general $\gamma > 0$, $\tan\theta = \gamma/2$.

> $\boldsymbol{U}$ and $\boldsymbol{V}$ have the same eigenvalues $\lambda_1, \lambda_2$ but different eigenvectors; $\boldsymbol{R}$ carries one set of principal axes onto the other. The polar decomposition is treated in Holzapfel (2000, Ch. 2) and Gurtin, Fried & Anand (2010); the numbers here were computed for these notes.

# STRAIN INVARIANTS AND VOLUME CHANGE

In three dimensions,

$$I_1 = \operatorname{tr} \boldsymbol{C}, \quad I_2 = \frac{1}{2}\left[(\operatorname{tr} \boldsymbol{C})^2 - \operatorname{tr}(\boldsymbol{C}^2)\right], \quad I_3 = \det \boldsymbol{C} = J^2.$$

**Raw invariants**

$I_1$ and $I_2$ respond to both shape change and volume change. $I_3$ measures volume change directly.

**Isochoric invariants**

Using $\bar{\boldsymbol{F}}_\chi = J^{-1/3}\boldsymbol{F}_\chi$ and $\bar{\boldsymbol{C}} = \bar{\boldsymbol{F}}_\chi{}^T \bar{\boldsymbol{F}}_\chi$,

$$\bar{I}_1 = \operatorname{tr} \bar{\boldsymbol{C}}, \qquad \bar{I}_2 = \tfrac{1}{2}[(\operatorname{tr} \bar{\boldsymbol{C}})^2 - \operatorname{tr}(\bar{\boldsymbol{C}}^2)].$$

# MODULE 2 WRAP-UP

- $\nabla_X \boldsymbol{d} = \boldsymbol{F}_\chi - \boldsymbol{I}$ is the displacement gradient.
- $\boldsymbol{C} = \boldsymbol{F}_\chi^T \boldsymbol{F}_\chi$ and $\boldsymbol{B} = \boldsymbol{F}_\chi \boldsymbol{F}_\chi^T$ encode stretch.
- $\boldsymbol{E} = \frac{1}{2}(\boldsymbol{C} - \boldsymbol{I})$ is a material finite-strain measure.
- $\boldsymbol{e} = \frac{1}{2}(\boldsymbol{I} - \boldsymbol{B}^{-1})$ is its spatial counterpart.
- $I_3 = J^2$. Modified invariants isolate isochoric deformation.

> Quick check: $J = 1$ implies $I_3 = 1$, but it does not imply $\boldsymbol{E} = \boldsymbol{0}$.

# MODULE 3: WHY SEVERAL STRESS MEASURES?

Fluid mechanics normally uses the Cauchy stress

$$\boldsymbol{\sigma}(\boldsymbol{x}, t)$$

in the current configuration. IBFE constitutive calculations usually use reference coordinates and the stresses

$$\boldsymbol{P}(\boldsymbol{X}, t) \qquad \text{and} \qquad \boldsymbol{S}(\boldsymbol{X}, t).$$

> Each stress measure pairs with a particular normal, area measure, or deformation rate. Conversions between them describe the same physical force or power.

# CAUCHY STRESS AND CURRENT TRACTION

For a current surface with outward unit normal $\boldsymbol{n}$,

$$\boxed{\boldsymbol{t}(\boldsymbol{n}) = \boldsymbol{\sigma}\,\boldsymbol{n}}.$$

The differential force acting on current area $da$ is

$$d\boldsymbol{f} = \boldsymbol{\sigma}\,\boldsymbol{n}\,da.$$

In components,

$$t_i = \sigma_{ij} n_j.$$

> The Cauchy stress maps a current normal to force per current area. Its diagonal entries describe normal stress and its off-diagonal entries describe shear stress in the chosen basis.

# TRACTION VECTORS ON COORDINATE PLANES

$$\boldsymbol{t}(\boldsymbol{n}) = \boldsymbol{\sigma}\boldsymbol{n}.$$

Choosing each coordinate normal selects one column of the Cauchy-stress matrix.

For example,

$$\boldsymbol{t}(\boldsymbol{e}_1) = \begin{bmatrix} \sigma_{11} \\ \sigma_{21} \\ \sigma_{31} \end{bmatrix}.$$

$$\mathbf{t}^{(\mathbf{e}_j)} = \boldsymbol{\sigma}\mathbf{e}_j, \qquad \begin{aligned} \mathbf{t}^{(\mathbf{e}_1)} &= \sigma_{11}\mathbf{e}_1 + \sigma_{21}\mathbf{e}_2 + \sigma_{31}\mathbf{e}_3 \\ \mathbf{t}^{(\mathbf{e}_2)} &= \sigma_{12}\mathbf{e}_1 + \sigma_{22}\mathbf{e}_2 + \sigma_{32}\mathbf{e}_3 \\ \mathbf{t}^{(\mathbf{e}_3)} &= \sigma_{13}\mathbf{e}_1 + \sigma_{23}\mathbf{e}_2 + \sigma_{33}\mathbf{e}_3 \end{aligned} \qquad (3.3.7)$$

Eqns. 3.3.6-7 are illustrated in Fig. 3.3.2.

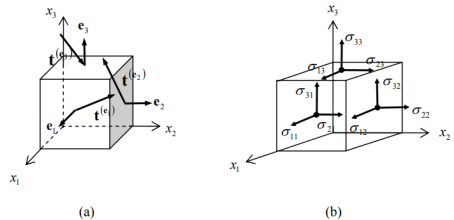



**Figure 3.3.2: traction acting on surfaces with normals in the coordinate directions; (a) traction vectors, (b) stress components**

Source: P. A. Kelly, *Solid Mechanics Part III: Foundations of Continuum Solid Mechanics*, §3.3, Fig. 3.3.2 (University of Auckland lecture notes, https://pkel015.connect.amazon.auckland.ac.nz/SolidMechanicsBooks/). The figure's equation numbers refer to those notes.

## FIRST PIOLA–KIRCHHOFF STRESS

The first Piola stress is a two-point tensor defined on $B_0$:

$$\boldsymbol{P}(\boldsymbol{X}, t) \in \mathbb{R}^{d\times d}.$$

For a reference surface with outward unit normal $\boldsymbol{N}$ and reference area $dA$,

$$\boxed{d\boldsymbol{f} = \boldsymbol{P}\boldsymbol{N}\, dA}.$$

Thus

$$\boldsymbol{t}_0(\boldsymbol{N}) = \boldsymbol{P}\boldsymbol{N}$$

is force per reference area, directed in the current configuration.

> $\boldsymbol{P}$ is generally nonsymmetric because its two indices refer to different configurations.

# MATCHING REFERENCE AND CURRENT SURFACE PATCHES

The same physical force can be described on corresponding patches:

$$d\boldsymbol{f} = \boldsymbol{\sigma n}\, da = \boldsymbol{PN}\, dA.$$

- $\boldsymbol{\sigma n}$: force per current area
- $\boldsymbol{PN}$: force per reference area

> The area vectors differ. Nanson's formula supplies the missing geometric transformation.

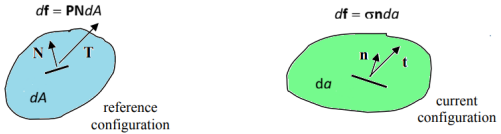



**Figure 3.5.1: Traction vectors**

Source: Kelly, *Solid Mechanics Part III*, §3.5, Fig. 3.5.1 (same notes). NB: the figure labels the reference traction $\boldsymbol{PN}$ as $\boldsymbol{T}$; in this course that quantity is $\boldsymbol{t}_0$, and $\boldsymbol{T}$ is reserved for the transmission force $-\boldsymbol{PN}$.

## NANSON'S FORMULA

Corresponding oriented area elements satisfy

$$\boxed{\boldsymbol{n}\, da = J\, \boldsymbol{F}_\chi^{-T} \boldsymbol{N}\, dA}.$$

**Reference area vector** $\boldsymbol{N}\, dA$ uses the material normal and reference area.

**Current area vector** $\boldsymbol{n}\, da$ uses the spatial normal and current area.

> In three dimensions, the identity follows by transforming two tangent vectors and using
>
> $$(\boldsymbol{F}_\chi \boldsymbol{a}) \times (\boldsymbol{F}_\chi \boldsymbol{b}) = J \boldsymbol{F}_\chi^{-T} (\boldsymbol{a} \times \boldsymbol{b}).$$

# PIOLA TRANSFORMATION

Start from equality of force on matching patches:

$$\boldsymbol{\sigma n}\, da = \boldsymbol{P N}\, dA.$$

Insert Nanson's formula:

$$\boldsymbol{\sigma}\left(J \boldsymbol{F}_{\chi}^{-T} \boldsymbol{N}\, dA\right) = \boldsymbol{P N}\, dA.$$

Because the relation holds for every $\boldsymbol{N}$,

$$\boxed{\boldsymbol{P} = J\, \boldsymbol{\sigma}\, \boldsymbol{F}_{\chi}^{-T}}.$$

Equivalently,

$$\boxed{\boldsymbol{\sigma} = J^{-1} \boldsymbol{P} \boldsymbol{F}_{\chi}^{T}}.$$

# SECOND PIOLA–KIRCHHOFF STRESS

Pull the remaining spatial index of $\boldsymbol{P}$ back to the reference configuration:

$$\boxed{\boldsymbol{S} = \boldsymbol{F}_\chi^{-1}\boldsymbol{P}}, \qquad \boxed{\boldsymbol{P} = \boldsymbol{F}_\chi \boldsymbol{S}}.$$

| Configuration | Energy pairing |
| --- | --- |
| $\boldsymbol{S}$ has two material indices and acts entirely in the reference description. | $\boldsymbol{P} : \dot{\boldsymbol{F}}_\chi = \frac{1}{2}\boldsymbol{S} : \dot{\boldsymbol{C}}.$ |

> If the Cauchy stress is symmetric, then $\boldsymbol{S}$ is symmetric.

# DIMENSIONAL, SYMMETRY, AND POWER CHECKS

| Measure | Area basis | Symmetry | Power pairing |
|---|---|---|---|
| $\boldsymbol{\sigma}$ | current | symmetric* | $J\boldsymbol{\sigma} : \boldsymbol{D}$ |
| $\boldsymbol{P}$ | reference | generally no | $\boldsymbol{P} : \dot{\boldsymbol{F}}_{\chi}$ |
| $\boldsymbol{S}$ | reference | symmetric* | $\frac{1}{2}\boldsymbol{S} : \dot{\boldsymbol{C}}$ |

$$\boldsymbol{P} : \dot{\boldsymbol{F}}_{\chi} = \frac{1}{2}\boldsymbol{S} : \dot{\boldsymbol{C}} = J\,\boldsymbol{\sigma} : \boldsymbol{D}, \qquad \boldsymbol{D} = \operatorname{sym}(\nabla_x \boldsymbol{u}).$$

*Symmetry follows for a classical nonpolar continuum satisfying angular-momentum balance.

# MODULE 3 WRAP-UP

- Current traction: $\boldsymbol{t} = \boldsymbol{\sigma n}$.
- Reference traction: $\boldsymbol{t}_0 = \boldsymbol{PN}$.
- Oriented areas:

$$\boldsymbol{n}\, da = J\boldsymbol{F}_\chi^{-T}\boldsymbol{N}\, dA.$$

- Stress conversions:

$$\boldsymbol{P} = J\boldsymbol{\sigma}\boldsymbol{F}_\chi^{-T}, \qquad \boldsymbol{\sigma} = J^{-1}\boldsymbol{P}\boldsymbol{F}_\chi^{T}, \qquad \boldsymbol{S} = \boldsymbol{F}_\chi^{-1}\boldsymbol{P}.$$

> The normal, area measure, and stress tensor must belong to matching configurations.

# IBFE NOTATION BRIDGE

| Object | Symbol here | Role in IBFE |
|---|---|---|
| Deformation gradient | $\boldsymbol{F}_\chi = \nabla_X \chi$ | structural kinematics |
| First Piola stress | $\boldsymbol{P}$ | constitutive response |
| Lagrangian force | $\boldsymbol{F}$ | weak structural force |
| Eulerian force | $\boldsymbol{f}$ | fluid momentum source |
| Cauchy stress | $\boldsymbol{\sigma}$ | current-configuration stress |

$$W(\boldsymbol{F}_\chi) \longrightarrow \boldsymbol{P} \longrightarrow \boldsymbol{F} \longrightarrow \boldsymbol{f}.$$

# MODULE 4: HYPERELASTICITY

A hyperelastic material stores energy according to a scalar density per reference volume:

$$W : \mathbb{R}^{d\times d} \to \mathbb{R}, \qquad \boldsymbol{F}_\chi \mapsto W(\boldsymbol{F}_\chi).$$

The first Piola stress follows by differentiation:

$$\boxed{\boldsymbol{P} = \frac{\partial W}{\partial \boldsymbol{F}_\chi}}.$$

> One scalar function encodes the constitutive response. Differentiating once gives stress. Differentiating again gives the material tangent used in nonlinear solvers.

## DERIVATIVE OF A SCALAR WITH RESPECT TO A MATRIX

The derivative has the same tensor shape as its matrix argument:

$$\left(\frac{\partial W}{\partial \boldsymbol{F}_\chi}\right)_{iI} := \frac{\partial W}{\partial (F_\chi)_{iI}}.$$

In three dimensions,

$$\frac{\partial W}{\partial \boldsymbol{F}_\chi} = \begin{bmatrix} \dfrac{\partial W}{\partial (F_\chi)_{11}} & \dfrac{\partial W}{\partial (F_\chi)_{12}} & \dfrac{\partial W}{\partial (F_\chi)_{13}} \\ \dfrac{\partial W}{\partial (F_\chi)_{21}} & \dfrac{\partial W}{\partial (F_\chi)_{22}} & \dfrac{\partial W}{\partial (F_\chi)_{23}} \\ \dfrac{\partial W}{\partial (F_\chi)_{31}} & \dfrac{\partial W}{\partial (F_\chi)_{32}} & \dfrac{\partial W}{\partial (F_\chi)_{33}} \end{bmatrix}.$$

> Shape check: $W$ is scalar, while $\partial W/\partial \boldsymbol{F}_\chi \in \mathbb{R}^{d\times d}$.

# DIRECTIONAL DERIVATIVE AND DOUBLE CONTRACTION

Perturb $\boldsymbol{F}_\chi$ in an arbitrary direction $\delta\boldsymbol{F}_\chi$:

$$\delta W := \frac{d}{d\varepsilon} W(\boldsymbol{F}_\chi + \varepsilon\,\delta\boldsymbol{F}_\chi)\Big|_{\varepsilon=0}.$$

The matrix derivative is defined by

$$\boxed{\delta W = \frac{\partial W}{\partial \boldsymbol{F}_\chi} : \delta\boldsymbol{F}_\chi = \boldsymbol{P} : \delta\boldsymbol{F}_\chi}.$$

The double contraction is the Frobenius inner product:

$$\boldsymbol{A} : \boldsymbol{B} = \sum_{i,I} A_{iI} B_{iI} = \operatorname{tr}(\boldsymbol{A}^T \boldsymbol{B}).$$

# OBJECTIVITY AND ISOTROPY

**Frame indifference**

For every proper orthogonal spatial rotation $\boldsymbol{Q}$,

$$W(\boldsymbol{Q}\boldsymbol{F}_\chi) = W(\boldsymbol{F}_\chi).$$

An objective energy may be written in terms of $\boldsymbol{C} = \boldsymbol{F}_\chi^T \boldsymbol{F}_\chi$.

**Material isotropy**

For every proper orthogonal material rotation $\boldsymbol{R}$,

$$W(\boldsymbol{F}_\chi \boldsymbol{R}) = W(\boldsymbol{F}_\chi).$$

An objective isotropic energy may be written using the invariants of $\boldsymbol{C}$.

> An anisotropic material can remain objective. Its energy also depends on material directions or structural tensors.

# COMPRESSIBLE AND INCOMPRESSIBLE RESPONSE

## Exact incompressibility

$$J = 1$$

is a constraint. A pressure-like Lagrange multiplier contributes

$$-p_s \, J \boldsymbol{F}_\chi^{-T}$$

to the first Piola stress.

## Nearly incompressible model

$$W(\boldsymbol{F}_\chi) = W_{\text{iso}}(\bar{\boldsymbol{F}}_\chi) + U(J)$$

uses a volumetric energy to discourage numerical volume change.

$$\dot{J} = J \, (\nabla_x \cdot \boldsymbol{u}) \circ \chi.$$

> At the continuum level, no slip and $\nabla_x \cdot \boldsymbol{u} = 0$ preserve the initial value $J(\boldsymbol{X}, 0)$. They give $J \equiv 1$ only when $J(\boldsymbol{X}, 0) = 1$. Discrete IBFE solutions can still develop local volume error.

# ISOCHORIC AND VOLUMETRIC DEFORMATION

In spatial dimension $d$, define

$$\boxed{\bar{\boldsymbol{F}}_\chi = J^{-1/d}\boldsymbol{F}_\chi}, \qquad \det \bar{\boldsymbol{F}}_\chi = 1.$$

Then

$$\bar{\boldsymbol{C}} = \bar{\boldsymbol{F}}_\chi{}^T \bar{\boldsymbol{F}}_\chi,$$

and the modified invariants of $\bar{\boldsymbol{C}}$ describe the isochoric response.

$\bar{\boldsymbol{F}}_\chi = J^{-1/2}\boldsymbol{F}_\chi$ in 2D, $\qquad$ $\bar{\boldsymbol{F}}_\chi = J^{-1/3}\boldsymbol{F}_\chi$ in 3D.

# ISOCHORIC AND VOLUMETRIC DEFORMATION (CONTINUED)

Equivalently, in 3D the deformation gradient (not the motion itself) splits multiplicatively into a volumetric (dilational) part $J^{1/3}\boldsymbol{I}$ and a distortional part $\bar{\boldsymbol{F}}_\chi$,

$$\boldsymbol{F}_\chi = (J^{1/3}\boldsymbol{I})\,\bar{\boldsymbol{F}}_\chi = J^{1/3}\bar{\boldsymbol{F}}_\chi, \qquad \bar{\boldsymbol{F}}_\chi = J^{-1/3}\boldsymbol{F}_\chi, \qquad \det(J^{1/3}\boldsymbol{I}) = J.$$

Consequently $\bar{\boldsymbol{C}} = J^{-2/3}\boldsymbol{C}$, so $\bar{I}_1 = J^{-2/3}I_1$ and $\bar{I}_2 = J^{-4/3}I_2$, while $\det\bar{\boldsymbol{C}} = 1$.

> Building the distortional energy from modified invariants prevents the shear response from also penalizing volume change.
>
> Griffith & Patankar (2020) introduce this split and the penalty $U(J)$ in §2.2.2; in §4.2 and Fig. 4 (summarizing Vadala-Roth, Rossi & Griffith 2018) they report that modified invariants together with volumetric stabilization give accuracy comparable to benchmark finite-element results in a torsion test of an incompressible Mooney–Rivlin beam similar to one in Bonet, Gil & Ortigosa (2015).

# VOLUMETRIC STABILIZATION

A common positive-$J$ choice is

$$U(J) = \frac{\kappa_{\mathrm{stab}}}{2}(\ln J)^2.$$

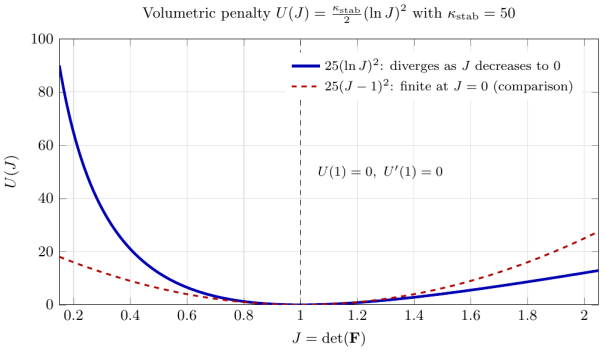
It satisfies

$$U(1) = 0, \qquad U'(1) = 0.$$

Unlike a quadratic $(J-1)^2$ penalty, $(\ln J)^2$ diverges as $J \to 0^+$: the barrier shown in the plot.

Its Piola-stress contribution is

$$\boldsymbol{P}_{\mathrm{vol}} = U'(J) J \boldsymbol{F}_\chi^{-T} = \kappa_{\mathrm{stab}} \ln J \, \boldsymbol{F}_\chi^{-T}.$$




Plot generated for these notes (pgfplots), $\kappa_{\mathrm{stab}} = 50$ (dimensionless); the axis label $\det(\mathbf{F})$ means $\det \boldsymbol{F}_\chi$.

In an incompressible IBFE model, $\kappa_{\mathrm{stab}}$ controls numerical volume error. It need not equal a physical bulk modulus.

# VOLUMETRIC STABILIZATION: DERIVATIVE CHECKS

For $U(J) = \frac{\kappa_{\text{stab}}}{2}(\ln J)^2$,

$$U'(J) = \kappa_{\text{stab}} \frac{\ln J}{J}, \qquad U''(J) = \kappa_{\text{stab}} \frac{1 - \ln J}{J^2}, \qquad U'(1) = 0, \quad U''(1) = \kappa_{\text{stab}}.$$

- $U''(1) = \kappa_{\text{stab}}$ is the small-strain volumetric stiffness, so $\kappa_{\text{stab}}$ plays the role of a bulk modulus near $J = 1$.
- The associated pressure is $p_{\text{vol}} = -U'(J) = -\kappa_{\text{stab}} \ln J / J$: positive (compressive) for $J < 1$, negative for $J > 1$.
- Using $\delta J = J \boldsymbol{F}_\chi^{-T} : \delta \boldsymbol{F}_\chi$ (derived later in this deck), $\delta U = U'(J)\, J \boldsymbol{F}_\chi^{-T} : \delta \boldsymbol{F}_\chi$, which gives $\boldsymbol{P}_{\text{vol}}$ on the previous slide.

> Griffith & Patankar (2020, §2.2.2) use this penalty, note that $U(1) = U'(1) = 0$ adds no energy or stress for isochoric motion, and interpret $\kappa_{\text{stab}}$ (their penalty coefficient) as a stabilization parameter because the immersed formulation already implies $J \equiv 1$.

# COMPRESSIBLE NEO-HOOKEAN ENERGY

One common three-dimensional model is

$$\boxed{W(\boldsymbol{F}_\chi) = \frac{\mu}{2}(I_1 - 3) - \mu \ln J + \frac{\lambda_{\mathrm{L}}}{2}(\ln J)^2},$$

where

$$I_1 = \operatorname{tr}(\boldsymbol{F}_\chi^T \boldsymbol{F}_\chi), \qquad J = \det \boldsymbol{F}_\chi > 0.$$

- $\mu$ is the infinitesimal shear modulus.
- $\lambda_{\mathrm{L}}$ is the first Lamé parameter.
- The infinitesimal bulk modulus is $K = \lambda_{\mathrm{L}} + 2\mu/3$.

> This distinction keeps a material Lamé parameter separate from the optional stabilization coefficient $\kappa_{\mathrm{stab}}$. In a (nearly) incompressible IBFE model, the $(\ln J)^2$ term of the neo-Hookean $W$ plays the role of $\kappa_{\mathrm{stab}}$ (details on the next slide).

Model as in Bonet & Wood (2008), hyperelasticity chapter (compressible neo-Hookean material); see also Holzapfel (2000), chapter on hyperelastic materials.

# LAMÉ PARAMETER VERSUS VOLUMETRIC STABILIZATION

Compare the last neo-Hookean term with the volumetric penalty introduced a few slides earlier:

$$\frac{\lambda_{\mathrm{L}}}{2}(\ln J)^2 \quad \longleftrightarrow \quad U(J) = \frac{\kappa_{\mathrm{stab}}}{2}(\ln J)^2.$$

- The forms are identical with $\kappa_{\mathrm{stab}} = \lambda_{\mathrm{L}}$, so in a nearly incompressible IBFE run a large $\lambda_{\mathrm{L}}$ acts as the volumetric penalty.
- It is not the whole volumetric response: $-\mu \ln J$ and the raw invariant $I_1$ also change with $J$. That is why the small-strain bulk modulus is $K = \lambda_{\mathrm{L}} + 2\mu/3$, not $\lambda_{\mathrm{L}}$.
- A cleanly decoupled alternative uses $W_{\mathrm{iso}}(\bar{\boldsymbol{F}}_\chi) + U(J)$, where only $U$ responds to volume change.

> When reporting parameters, state whether $\lambda_{\mathrm{L}}$ is a measured material constant or a numerical stabilization choice; the two interpretations call for very different values.

# NEO-HOOKEAN FIRST PIOLA STRESS

For the compressible neo-Hookean energy $W = \frac{\mu}{2}(I_1 - 3) - \mu \ln J + \frac{\lambda_L}{2}(\ln J)^2$ introduced above,

$$\boxed{\boldsymbol{P} = \mu \left(\boldsymbol{F}_\chi - \boldsymbol{F}_\chi^{-T}\right) + \lambda_L \ln J \, \boldsymbol{F}_\chi^{-T}}.$$

**Reference state**

At $\boldsymbol{F}_\chi = \boldsymbol{I}$,

$$J = 1, \qquad \boldsymbol{P} = \boldsymbol{0}.$$

**Isochoric state**

If $J = 1$,

$$\boldsymbol{P} = \mu(\boldsymbol{F}_\chi - \boldsymbol{F}_\chi^{-T}).$$

> The final section derives this expression term by term.

# NEO-HOOKEAN STRESS CONVERSIONS

Once $\boldsymbol{P}$ is known,

$$\boldsymbol{S} = \boldsymbol{F}_\chi^{-1}\boldsymbol{P}, \qquad \boldsymbol{\sigma} = J^{-1}\boldsymbol{P}\boldsymbol{F}_\chi^{T}.$$

For the compressible neo-Hookean model,

$$\boxed{\boldsymbol{\sigma} = \frac{\mu}{J}(\boldsymbol{B} - \boldsymbol{I}) + \frac{\lambda_{\mathrm{L}} \ln J}{J}\boldsymbol{I}}.$$

> $\boldsymbol{\sigma}$ is symmetric because $\boldsymbol{B}$ and $\boldsymbol{I}$ are symmetric. The same constitutive response appears different because each stress uses a different configuration and area measure.

# BEYOND NEO-HOOKEAN: OTHER ISOTROPIC HYPERELASTIC MODELS

Each model below specifies only $W$; the pipeline $W \to \boldsymbol{P} = \partial W/\partial \boldsymbol{F}_\chi \to \boldsymbol{F} \to \boldsymbol{f}$ is unchanged. Bars denote modified invariants, and $U(J)$ is a volumetric penalty as before.

## Mooney–Rivlin (rubber-like solids)

$$W = c_1(\bar{I}_1 - 3) + c_2(\bar{I}_2 - 3) + U(J).$$

Small-strain shear modulus $\mu = 2(c_1 + c_2)$; $c_2 = 0$ recovers a (modified-invariant) neo-Hookean solid. This is the material in the Griffith & Patankar (2020) torsion test.

## Fung-type exponential (soft tissue)

$$W = \frac{a}{2b}\left(e^{\,b(\bar{I}_1 - 3)} - 1\right) + U(J).$$

Exponential stiffening with stretch, typical of collagenous tissue; $\mu = a$ at small strain, and $b \to 0$ recovers neo-Hookean with $\mu = a$.

> Check for any candidate $W$: $\boldsymbol{P}(\boldsymbol{I}) = \boldsymbol{0}$ (stress-free reference state), objectivity (depends on $\boldsymbol{F}_\chi$ only through $\boldsymbol{C}$), and the correct small-strain shear modulus.

Mooney (1940), Rivlin (1948); exponential form after Fung (1993) and Demiray (1972). See the reference list.

# FIBER-REINFORCED MODELS (HOLZAPFEL AND CO-WORKERS)

Anisotropy enters through pseudo-invariants built from reference fiber directions $\boldsymbol{f}_0$, $\boldsymbol{s}_0$:

$$I_{4f} = \boldsymbol{f}_0 \cdot \boldsymbol{C}\boldsymbol{f}_0 = \lambda_f^2, \qquad I_{8fs} = \boldsymbol{f}_0 \cdot \boldsymbol{C}\boldsymbol{s}_0.$$

## Arterial wall (HGO, 2000)

$$W = \frac{\mu}{2}(\bar{I}_1 - 3) + \sum_{i=4,6} \frac{k_1}{2k_2}\left(e^{\,k_2(\bar{I}_i - 1)^2} - 1\right) + U(J),$$

two collagen families $I_4, I_6$; the fiber terms are usually switched on only in tension ($\bar{I}_i > 1$).

## Myocardium (HO, 2009)

$$W = \frac{a}{2b}\, e^{\,b(I_1 - 3)} + \sum_{i=f,s} \frac{a_i}{2b_i}\left(e^{\,b_i(I_{4i} - 1)^2} - 1\right) + \frac{a_{fs}}{2b_{fs}}\left(e^{\,b_{fs} I_{8fs}^2} - 1\right)$$

(up to an additive constant), plus a volumetric term; fiber and sheet terms act in tension.

> These are the passive laws typically combined with an active fiber stress (deck 4-3) in cardiac IBFE models; the derivative of an $I_4$ term is $\partial I_{4f}/\partial \boldsymbol{F}_\chi = 2\boldsymbol{F}_\chi \boldsymbol{f}_0 \otimes \boldsymbol{f}_0$.

Holzapfel, Gasser & Ogden (2000); Holzapfel & Ogden (2009). Constants and tension switches vary between implementations; check the source before reuse.

# IBFE FORCE PIPELINE

1. Evaluate $\boldsymbol{F}_{\chi h} = \nabla_X \chi_h$, $J_h$, and $\boldsymbol{P}_h = \partial W / \partial \boldsymbol{F}_\chi$ at structural quadrature points.
2. Define the unified Lagrangian force by negative virtual work:

$$\int_{B_0} \boldsymbol{F}_h \cdot \boldsymbol{V}_h \, dV = - \int_{B_0} \boldsymbol{P}_h : \nabla_X \boldsymbol{V}_h \, dV.$$

3. Spread $\boldsymbol{F}_h$ to the Eulerian grid with a regularized delta kernel.
4. Advance the incompressible fluid equations.
5. Restrict the Eulerian velocity back to the structure and update $\chi_h$.

> The negative sign makes internal elastic force oppose increasing stored energy. Prescribed body forces or boundary tractions add their own virtual-work terms.

# FINITE-ELEMENT INTERPOLATION OF MOTION AND STRESS

With structural basis functions $\{\phi_\ell(\boldsymbol{X})\}$,

$$\chi_h(\boldsymbol{X}, t) = \sum_{\ell=1}^{M} \chi_\ell(t)\phi_\ell(\boldsymbol{X}),$$

$$\boldsymbol{F}_{\chi_h}(\boldsymbol{X}, t) = \sum_{\ell=1}^{M} \chi_\ell(t) \otimes \nabla_X \phi_\ell(\boldsymbol{X}).$$

- $C^0$ interpolation makes $\chi_h$ continuous.
- $\boldsymbol{F}_{\chi_h}$ and $\boldsymbol{P}_h$ are generally discontinuous across element boundaries.
- Constitutive stress and interaction forces are evaluated at element quadrature points.
- Interaction quadrature may be refined independently of structural nodes and helps prevent leakage when adequately resolved.

# MODULE 4 WRAP-UP

- Hyperelasticity defines $\boldsymbol{P} = \partial W / \partial \boldsymbol{F}_\chi$.
- The identity $\delta W = \boldsymbol{P} : \delta \boldsymbol{F}_\chi$ defines the matrix derivative.
- Objectivity and isotropy impose different constitutive requirements.
- Modified invariants and $U(J)$ separate shape and volume response.
- The weak structural force carries a negative virtual-work sign.
- FE quadrature connects piecewise constitutive fields to the Eulerian grid.

> Next, one homogeneous deformation will pass through the entire chain from $\chi$ to $\boldsymbol{\sigma}$.

# WORKED EXAMPLE: UNIAXIAL STRETCH (PRESCRIBED AXIAL DEFORMATION)

Consider the three-dimensional motion

$$\chi(\boldsymbol{X}) = \begin{bmatrix} \alpha X_1 \\ X_2 \\ X_3 \end{bmatrix}, \qquad \alpha > 0.$$

- Material points stretch by the factor $\alpha$ along $X_1$.
- The transverse coordinates remain unchanged.
- There is no shear.

> Throughout the course this example is called *uniaxial stretch (prescribed axial deformation)*; it is sometimes also called uniaxial strain. It is not automatically a uniaxial-stress test because the lateral stretches are fixed.

## AXIAL DEFORMATION GRADIENT AND JACOBIAN

Differentiate each component of the motion:

$$\boldsymbol{F}_\chi = \nabla_X \chi = \begin{bmatrix} \dfrac{\partial \chi_1}{\partial X_1} & \dfrac{\partial \chi_1}{\partial X_2} & \dfrac{\partial \chi_1}{\partial X_3} \\ \dfrac{\partial \chi_2}{\partial X_1} & \dfrac{\partial \chi_2}{\partial X_2} & \dfrac{\partial \chi_2}{\partial X_3} \\ \dfrac{\partial \chi_3}{\partial X_1} & \dfrac{\partial \chi_3}{\partial X_2} & \dfrac{\partial \chi_3}{\partial X_3} \end{bmatrix} = \operatorname{diag}(\alpha, 1, 1).$$

Therefore

$$\boxed{J = \det \boldsymbol{F}_\chi = \alpha}.$$

> The prescribed deformation changes volume whenever $\alpha \neq 1$.

# AXIAL CAUCHY–GREEN TENSOR AND GREEN STRAIN

Because $\boldsymbol{F}_\chi$ is diagonal,

$$\boldsymbol{C} = \boldsymbol{F}_\chi^T \boldsymbol{F}_\chi = \operatorname{diag}(\alpha^2, 1, 1).$$

Then

$$\boldsymbol{E} = \frac{1}{2}(\boldsymbol{C} - \boldsymbol{I}) = \operatorname{diag}\left(\frac{\alpha^2 - 1}{2}, 0, 0\right).$$

$$E_{11} = \frac{\alpha^2 - 1}{2}. \qquad E_{22} = E_{33} = 0.$$

> For $\alpha = 1 + \varepsilon$ with $|\varepsilon| \ll 1$, $E_{11} = \varepsilon + O(\varepsilon^2)$.

## AXIAL-DEFORMATION INVARIANTS

For

$$\boldsymbol{C} = \operatorname{diag}(\alpha^2, 1, 1),$$

the first and third invariants are

$$I_1 = \operatorname{tr} \boldsymbol{C} = \alpha^2 + 2,$$

$$I_3 = \det \boldsymbol{C} = \alpha^2 = J^2.$$

The second invariant is

$$I_2 = 1 + 2\alpha^2.$$

> $I_3$ directly records volume change. The raw invariants $I_1$ and $I_2$ reflect both the axial shape change and the accompanying volume change.

## CONSTITUTIVE CHOICE FOR THE AXIAL EXAMPLE

Use the compressible neo-Hookean energy

$$W(\boldsymbol{F}_\chi) = \frac{\mu}{2}(I_1 - 3) - \mu \ln J + \frac{\lambda_{\mathrm{L}}}{2}(\ln J)^2.$$

For the prescribed axial deformation,

$$I_1 = \alpha^2 + 2, \qquad J = \alpha,$$

so

$$W(\alpha) = \frac{\mu}{2}(\alpha^2 - 1) - \mu \ln \alpha + \frac{\lambda_{\mathrm{L}}}{2}(\ln \alpha)^2.$$

> The stress still follows by differentiating the full matrix-valued argument $\boldsymbol{F}_\chi$, not only the reduced scalar expression $W(\alpha)$.

# FIRST PIOLA STRESS FOR THE AXIAL DEFORMATION

Use

$$\boldsymbol{P} = \mu(\boldsymbol{F}_\chi - \boldsymbol{F}_\chi^{-T}) + \lambda_{\mathrm{L}} \ln J \, \boldsymbol{F}_\chi^{-T}.$$

Here

$$\boldsymbol{F}_\chi^{-T} = \mathrm{diag}(\alpha^{-1}, 1, 1), \qquad \ln J = \ln \alpha.$$

Therefore

$$\boxed{\boldsymbol{P} = \mathrm{diag}\left(\mu(\alpha - \alpha^{-1}) + \lambda_{\mathrm{L}} \frac{\ln \alpha}{\alpha}, \lambda_{\mathrm{L}} \ln \alpha, \lambda_{\mathrm{L}} \ln \alpha\right).}$$

> Fixing both lateral stretches to one generally produces nonzero transverse stress.

# CAUCHY STRESS FOR THE AXIAL DEFORMATION

Use the Piola transformation:

$$\boldsymbol{\sigma} = \frac{1}{J}\boldsymbol{P}\boldsymbol{F}_{\chi}^{T}.$$

Since $J = \alpha$ and both tensors are diagonal,

$$\boldsymbol{\sigma} = \operatorname{diag}\left(P_{11}, \frac{P_{22}}{\alpha}, \frac{P_{33}}{\alpha}\right).$$

Thus

$$\sigma_{22} = \sigma_{33} = \frac{\lambda_{\mathrm{L}} \ln \alpha}{\alpha}.$$

> A traction-free uniaxial-stress test must allow unknown lateral stretches and solve for them from $\sigma_{22} = \sigma_{33} = 0$.

# CONTRAST: WHAT A UNIAXIAL-STRESS TEST WOULD GIVE

Let the lateral stretches be unknown: $\boldsymbol{F}_\chi = \mathrm{diag}(\alpha, \lambda_\perp, \lambda_\perp)$, $J = \alpha\lambda_\perp^2$, $\boldsymbol{B} = \mathrm{diag}(\alpha^2, \lambda_\perp^2, \lambda_\perp^2)$.

## Compressible neo-Hookean

From $\boldsymbol{\sigma} = \frac{\mu}{J}(\boldsymbol{B} - \boldsymbol{I}) + \frac{\lambda_\mathrm{L} \ln J}{J}\boldsymbol{I}$, the condition $\sigma_{22} = 0$ reads

$$\mu(\lambda_\perp^2 - 1) + \lambda_\mathrm{L} \ln(\alpha\lambda_\perp^2) = 0,$$

a scalar nonlinear equation for $\lambda_\perp(\alpha)$ (solve by Newton's method).

## Incompressible neo-Hookean

With $J = 1$, $\lambda_\perp = \alpha^{-1/2}$ and $\boldsymbol{\sigma} = \mu\boldsymbol{B} - p\boldsymbol{I}$. Then $\sigma_{22} = 0$ gives $p = \mu/\alpha$ and

$$\sigma_{11} = \mu\left(\alpha^2 - \frac{1}{\alpha}\right).$$

> Same material, different boundary conditions: prescribing $\lambda_\perp = 1$ (this worked example) gives nonzero lateral stress; leaving the lateral faces traction free gives lateral contraction instead.

# AXIAL-EXAMPLE WRAP-UP

$$\chi(\boldsymbol{X}) \longrightarrow \boldsymbol{F}_\chi = \operatorname{diag}(\alpha, 1, 1) \longrightarrow J = \alpha$$

$$\longrightarrow \boldsymbol{C} = \operatorname{diag}(\alpha^2, 1, 1) \longrightarrow \boldsymbol{E} = \frac{1}{2}\operatorname{diag}(\alpha^2 - 1, 0, 0).$$

After choosing $W$,

$$W(\boldsymbol{F}_\chi) \longrightarrow \boldsymbol{P} = \frac{\partial W}{\partial \boldsymbol{F}_\chi} \longrightarrow \boldsymbol{\sigma} = J^{-1}\boldsymbol{P}\boldsymbol{F}_\chi^T.$$

> Every arrow can be checked independently. Homogeneous deformations are useful unit tests for constitutive implementations.

# DERIVATION GOAL

Start with

$$W(\boldsymbol{F}_\chi) = \frac{\mu}{2}(I_1 - 3) - \mu \ln J + \frac{\lambda_{\mathrm{L}}}{2}(\ln J)^2.$$

We will compute the first variation and match

$$\delta W = \boldsymbol{P} : \delta \boldsymbol{F}_\chi.$$

Two identities drive the calculation:

$$\delta I_1 = 2\boldsymbol{F}_\chi : \delta \boldsymbol{F}_\chi, \qquad \delta(\ln J) = \boldsymbol{F}_\chi^{-T} : \delta \boldsymbol{F}_\chi.$$

> The next two slides derive these identities before the terms are combined.

## STEP 1: VARIATION OF $I_1$

Because

$$I_1 = \mathrm{tr}(\boldsymbol{F}_\chi^T \boldsymbol{F}_\chi) = \sum_{i,I} (F_\chi)_{iI}^2,$$

$$\delta I_1 = 2 \sum_{i,I} (F_\chi)_{iI}\, \delta(F_\chi)_{iI} = \boxed{2\boldsymbol{F}_\chi : \delta \boldsymbol{F}_\chi}.$$

Therefore

$$\delta \left[ \frac{\mu}{2} (I_1 - 3) \right] = \mu \boldsymbol{F}_\chi : \delta \boldsymbol{F}_\chi.$$

> This term contributes $\mu \boldsymbol{F}_\chi$ to $\boldsymbol{P}$.

# STEP 2: VARIATION OF $\ln J$

Jacobi's determinant identity gives

$$\delta J = J\, \boldsymbol{F}_\chi^{-T} : \delta \boldsymbol{F}_\chi.$$

The chain rule then gives

$$\boxed{\delta(\ln J) = \frac{\delta J}{J} = \boldsymbol{F}_\chi^{-T} : \delta \boldsymbol{F}_\chi}.$$

Consequently,

$$\delta(-\mu \ln J) = -\mu \boldsymbol{F}_\chi^{-T} : \delta \boldsymbol{F}_\chi,$$

$$\delta\left[\frac{\lambda_{\mathrm{L}}}{2}(\ln J)^2\right] = \lambda_{\mathrm{L}} \ln J\, \boldsymbol{F}_\chi^{-T} : \delta \boldsymbol{F}_\chi.$$

# STEP 3: COMBINE THE STRESS CONTRIBUTIONS

Collect the coefficients of $\delta \boldsymbol{F}_\chi$:

$$\delta W = \left[\mu \boldsymbol{F}_\chi - \mu \boldsymbol{F}_\chi^{-T} + \lambda_{\mathrm{L}} \ln J \, \boldsymbol{F}_\chi^{-T}\right] : \delta \boldsymbol{F}_\chi.$$

Matching $\delta W = \boldsymbol{P} : \delta \boldsymbol{F}_\chi$ gives

$$\boxed{\boldsymbol{P} = \mu(\boldsymbol{F}_\chi - \boldsymbol{F}_\chi^{-T}) + \lambda_{\mathrm{L}} \ln J \, \boldsymbol{F}_\chi^{-T}}.$$

$$\boldsymbol{F}_\chi = \boldsymbol{I} \quad \Rightarrow \quad \boldsymbol{P} = \boldsymbol{0}. \qquad\qquad J = 1 \quad \Rightarrow \quad \boldsymbol{P} = \mu(\boldsymbol{F}_\chi - \boldsymbol{F}_\chi^{-T}).$$

# MICRO-CHECK: SMALL-STRAIN LIMIT

Let

$$\boldsymbol{F}_\chi = \boldsymbol{I} + \boldsymbol{H}, \qquad \|\boldsymbol{H}\| \ll 1, \qquad \boldsymbol{\varepsilon} = \operatorname{sym} \boldsymbol{H}.$$

To first order,

$$\boldsymbol{F}_\chi^{-T} \approx \boldsymbol{I} - \boldsymbol{H}^T, \qquad \ln J \approx \operatorname{tr} \boldsymbol{H}.$$

Substitution gives

$$\boldsymbol{P} \approx \mu(\boldsymbol{H} + \boldsymbol{H}^T) + \lambda_{\mathrm{L}} \operatorname{tr}(\boldsymbol{H})\boldsymbol{I} = \boxed{2\mu\boldsymbol{\varepsilon} + \lambda_{\mathrm{L}} \operatorname{tr}(\boldsymbol{\varepsilon})\boldsymbol{I}}.$$

> At first order, $\boldsymbol{P}$ and $\boldsymbol{\sigma}$ coincide. The model reduces to isotropic linear elasticity.

# MICRO-CHECK: AXIAL-DEFORMATION PLUG-IN

Use

$$\boldsymbol{F}_\chi = \mathrm{diag}(\alpha, 1, 1), \quad J = \alpha, \quad \boldsymbol{F}_\chi^{-T} = \mathrm{diag}(\alpha^{-1}, 1, 1).$$

**Task**
Insert these expressions into

$$\boldsymbol{P} = \mu(\boldsymbol{F}_\chi - \boldsymbol{F}_\chi^{-T}) + \lambda_{\mathrm{L}} \ln J \, \boldsymbol{F}_\chi^{-T}$$

and compute $P_{11}, P_{22}, P_{33}$.

$$P_{11} = \mu(\alpha - \alpha^{-1}) + \lambda_{\mathrm{L}} \frac{\ln \alpha}{\alpha}, \qquad P_{22} = P_{33} = \lambda_{\mathrm{L}} \ln \alpha.$$

These lateral stresses vanish only for $\alpha = 1$ under the prescribed lateral stretches.

# REFERENCES AND FURTHER READING

**Primary IB and IBFE references**


- B. E. Griffith and X. Luo, *Hybrid finite difference/finite element immersed boundary method*, Int. J. Numer. Methods Biomed. Eng. 33 (2017), e2888. doi:10.1002/cnm.2888
- B. E. Griffith and N. A. Patankar, *Immersed methods for fluid–structure interaction*, Annu. Rev. Fluid Mech. 52 (2020), 421–448. doi:10.1146/annurev-fluid-010719-060228
- C. S. Peskin, *The immersed boundary method*, Acta Numerica 11 (2002), 479–517. doi:10.1017/S0962492902000077


**Continuum mechanics**


- G. A. Holzapfel, *Nonlinear Solid Mechanics*, Wiley, 2000.
- J. Bonet and R. D. Wood, *Nonlinear Continuum Mechanics for Finite Element Analysis*, Cambridge, 2008.


**Open resources**

- Ajeet Kumar solid-mechanics notes
- NPTEL continuum-mechanics course
- IBAMR documentation and source

# REFERENCES AND FURTHER READING (CONTINUED)


**Further sources drawn on in this deck**

- P. A. Kelly, *Solid Mechanics Part III: Foundations of Continuum Solid Mechanics*, lecture notes, University of Auckland, §3.3 (Fig. 3.3.2) and §3.5 (Fig. 3.5.1), reproduced on the traction slides.
  https://pkel015.connect.amazon.auckland.ac.nz/SolidMechanicsBooks/
- J. Bonet, A. J. Gil, and R. Ortigosa, *A computational framework for polyconvex large strain elasticity*, Comput. Methods Appl. Mech. Engrg. 283 (2015), 1061–1094. doi:10.1016/j.cma.2014.10.002
- B. Vadala-Roth, S. Rossi, and B. E. Griffith, *Stabilization approaches for the hyperelastic immersed boundary method for problems of large-deformation incompressible elasticity*, arXiv preprint (2018), as cited by Griffith & Patankar (2020); later published in Comput. Methods Appl. Mech. Engrg. (2020).
- D. Boffi, L. Gastaldi, L. Heltai, and C. S. Peskin, *On the hyper-elastic formulation of the immersed boundary method*, Comput. Methods Appl. Mech. Engrg. 197 (2008), 2210–2231.

**Further reading**

- M. E. Gurtin, E. Fried, and L. Anand, *The Mechanics and Thermodynamics of Continua*, Cambridge, 2010 (polar decomposition, frame indifference).
- A. F. Bower, *Applied Mechanics of Solids*, CRC Press, 2010; https://solidmechanics.org.


**Figure credits**

- ref-current, def-tensor, Green-Lagrange: schematics from the instructor's course notes.
- $U(J)$ plot: generated for these notes (pgfplots).
- Traction figures: Kelly (above).

# REFERENCES: ADDITIONAL CONSTITUTIVE MODELS


- M. Mooney, A theory of large elastic deformation, *J. Appl. Phys.* 11 (1940) 582–592.
- R. S. Rivlin, Large elastic deformations of isotropic materials IV: further developments of the general theory, *Phil. Trans. R. Soc. Lond. A* 241 (1948) 379–397.
- Y. C. Fung, *Biomechanics: Mechanical Properties of Living Tissues*, 2nd ed., Springer, 1993.
- H. Demiray, A note on the elasticity of soft biological tissues, *J. Biomech.* 5 (1972) 309–311.
- G. A. Holzapfel, T. C. Gasser, and R. W. Ogden, A new constitutive framework for arterial wall mechanics and a comparative study of material models, *J. Elasticity* 61 (2000) 1–48.
- G. A. Holzapfel and R. W. Ogden, Constitutive modelling of passive myocardium: a structurally based framework for material characterization, *Phil. Trans. R. Soc. A* 367 (2009) 3445–3475.


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